

DATA SCIENCE – PLACEMENT DATASET ANALYSIS

1. Replace the NaN values with correct value. And justify why you have chosen the same.

Only the **salary column** had **67 missing values** in the dataset. Since missing salary indicates **students who were not placed**, the NaN values were replaced with 0. This is the most appropriate choice because it correctly represents “no placement” without affecting the dataset distribution.

2. How many of them are not placed?

The dataset analysis shows that **67 students are not placed**. This value was obtained from the status column using value counts, where “Not Placed” entries were filtered and counted.

3. Find the reason for non placement from the dataset?

From the dataset analysis, students who are not placed generally have **lower average scores in SSC, HSC, degree percentage, MBA percentage, and E-test scores** compared to placed students. Also, students without **work experience** show a higher rate of non-placement. Therefore, poor academic performance and lack of experience are the main reasons for non-placement in this dataset.

4. What kind of relation between salary and mba_p?

The correlation coefficient between salary and MBA percentage is **0.1398**, indicating a **weak positive (directly proportional) relationship**. As MBA percentage increases, salary tends to increase slightly, but the influence is very weak in this dataset.

5. Which specialization is getting minimum salary?

From the analysis, both **Mkt&Fin** and **Mkt&HR specialisations** have the same minimum salary of **0**. This indicates that in both groups, some students were not placed, resulting in zero salary values.

6. How many of them getting above 500000 salary?

From the dataset analysis, **3 students are getting a salary above 500000**. This was identified by filtering the salary column with the condition greater than 500000.

7. Test the Analysis of Variance between etest_p and mba_p at significance level 5%. (Make decision using Hypothesis Testing)

```
F_onewayResult(statistic=np.float64(98.64487057324706), pvalue=np.float64(4.672547689133573e-21))
```

Since the **p-value (4.67×10^{-21})** is less than **0.05**

There is a **significant difference** between **etest_p** and **mba_p**.

So, we **reject the Null Hypothesis** and accept the **Alternate Hypothesis**.

8. Test the similarity between the degree_t(Sci&Tech) and specialisation(Mkt&HR) with respect to salary at significance level of 5%. (Make decision using Hypothesis Testing)

```
TtestResult(statistic=np.float64(2.692041243555374), pvalue=np.float64(0.007897969943471179), df=np.float64(152.0))
```

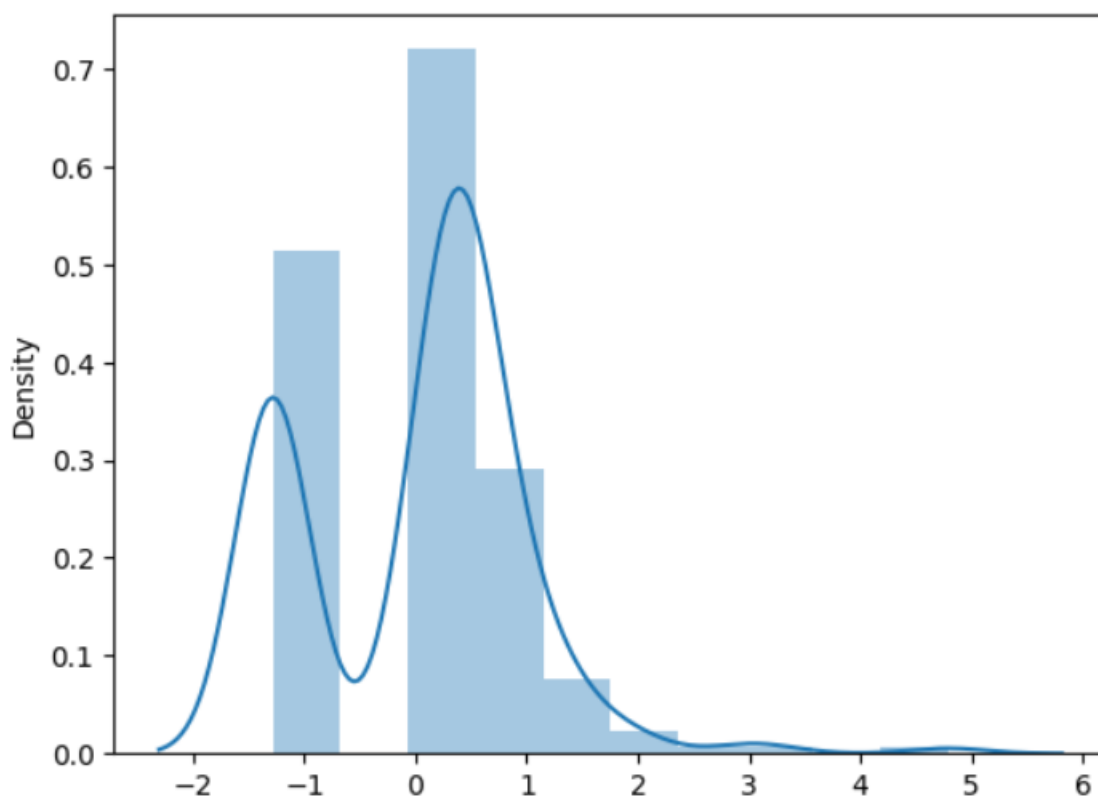
Since **p-value (0.0079)** is less than **0.05**, we reject the null hypothesis.

There is a **significant difference in salary between Sci&Tech and Mkt&HR groups**, so we accept the alternate hypothesis.

9.Convert the normal distribution to standard normal distribution for salary column

	salary	salary_zscore
0	270000.0	0.460636
1	200000.0	0.008384
2	250000.0	0.331421
3	0.0	-1.283765
4	425000.0	1.462051

The salary column was transformed into a standard normal distribution using Z-score normalization. After transformation, the data has a mean approximately equal to 0 and a standard deviation equal to 1.



The histogram shows a near bell-shaped curve, indicating that the values are properly standardized. The Z-score table confirms that salary values are now scaled based on their deviation from the mean, making all values comparable on a common scale and improving machine learning model performance.

10.What is the probability Density Function of the salary range from700000 to 90000?

The probability value **0.000597** indicates that only a very small portion of the data lies between 700000 and 900000 salary range, meaning this range is rare in the dataset.

11. Test the similarity between the degree_t(Sci&Tech) with respect to etest_p and mba_p at significance level of 5%. (Make decision using Hypothesis Testing)

```
TtestResult(statistic=np.float64(5.0049844583693615), pvalue=np.float64(5.517920600505392e-06), df=np.int64(58))
```

Since the p-value (5.52×10^{-6}) is less than **0.05**, there is a significant difference between **etest_p** and **mba_p** scores of Sci&Tech students.

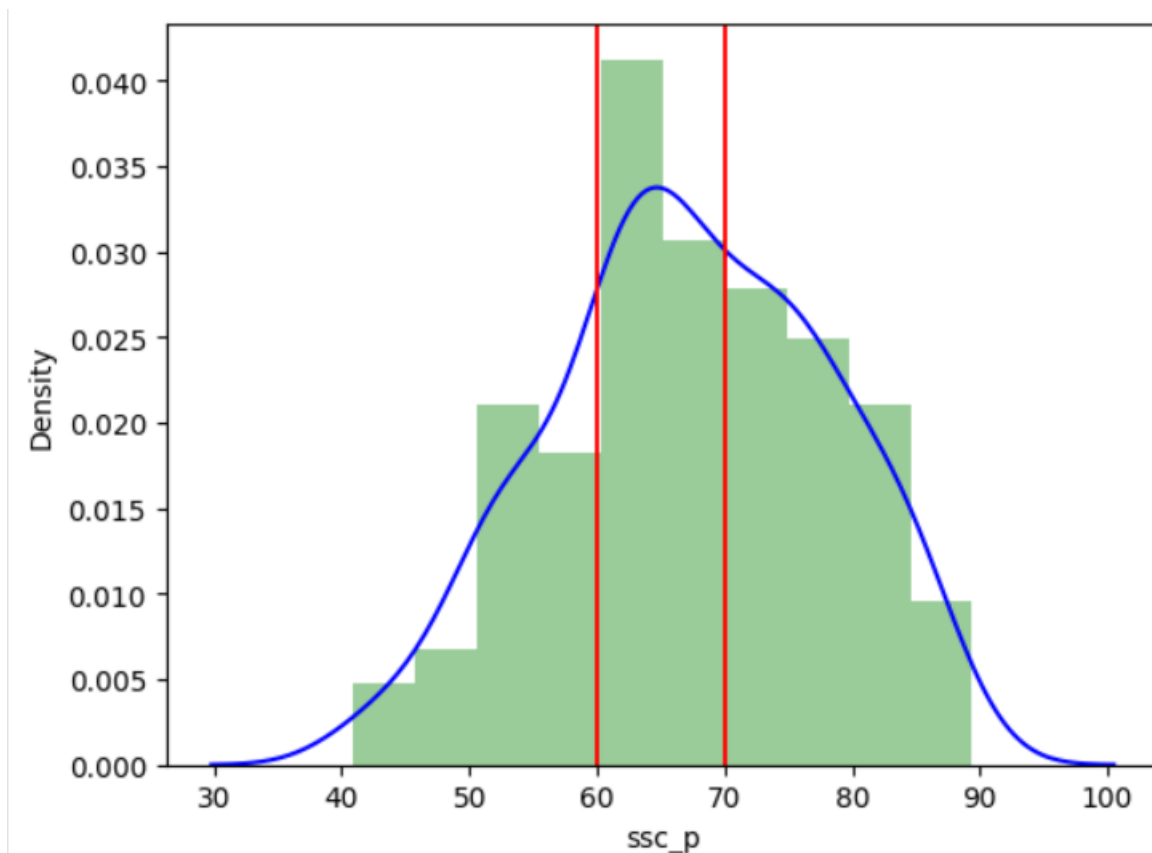
Therefore, we **reject the Null Hypothesis** and **accept the Alternative Hypothesis**.

12. Which parameter is highly correlated with salary?

salary	1.000000
ssc_p	0.538090
hsc_p	0.452569
degree_p	0.408371
etest_p	0.186988
mba_p	0.139823

ssc_p is highly correlated with salary, with a correlation coefficient of **0.538**, indicating a moderate positive relationship with salary.

13. Plot any useful graph and explain it?



The probability density graph of **ssc_p** shows a mean of **67.303** and a standard deviation of **10.827**. The range between **60 and 70** contains a probability area of **0.345 (34.5%)**, indicating that a significant portion of the data falls within this interval. Most values are concentrated around the mean, forming an approximately normal distribution.